STAT/MA 41600

In-Class Problem Set #32: October 30, 2026

1. At a haunted house, the time until the next vampire appears (from the East Parlor) is an exponential random variable with mean 45 seconds. The time until the next werewolf appears (from the West Parlor) is an exponential random variable with mean 40 seconds. These two random variables are independent. What is the probability that the next werewolf appears before the next vampire?

2. There are three mice waiting to emerge from three separate holes in a rickety old wooden barn, just after midnight. The time (after midnight) until each of the three mice run out of their holes are (respectively) X_1, X_2, X_3, three independent exponential random variables, with respective means of 2 minutes, 2.5 minutes, and 3 minutes. What is the probability that X_3 is larger than both X_1 and X_2, i.e., find $P(X_3 \geq \max(X_1, X_2))$.

3. Same situation as Question **2**: What is the probability that all three of the X_is are bigger than 2.25 minutes, i.e., $P(2.25 \leq \min(X_1, X_2, X_3))$?

4. At a corn maze, a helpful person shines a big flashlight into the corn maze at random intervals. The time (in minutes) until she shines the next flashlight is a continuous random variable that is uniformly distributed on $[0, 2]$. Independent of the flashlight, the time until the next zombie runs through the corn maze is an exponential random variable with mean 1.4 minutes. Calculate the probability that a zombie runs through the corn maze before the helpful person next shines their flashlight.

5. There are 4 available types of witches' brew (bubbling, smoky, acidic, and slimy). Upon drinking the potions, each person will begin to feel the effects of the potion in an exponential time with mean 2 minutes. Suppose that 4 people each simultaneously drink one of the witches' brews, and their times to feel the effects of the potions are independent. Find the probability that, after 2.5 minutes, at least 3 out of the 4 people begin to feel the effects of the potion.